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Ulrich ideals in numerical and good semigroups

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Value semigroups and good semigroups - Catania 2026

1. Ulrich modules and ideals

Let $(R, \mathfrak{m})$ be a local (or graded) Cohen-Macaulay ring. A Cohen-Macaulay R -module M is a **Maximal Cohen-Macaulay module** if $\text{depth}_{\mathfrak{m}}(M) = \dim(R)$.

Let $\nu(M)$, $e(M)$ be respectively the number of generators and the multiplicity of M .

Then $\nu(M) \leq e(M)$ and, if the equality holds M is said to be an **Ulrich module** (Herzog-Kühl, 1987) or, alternatively a Maximally generated Maximal Cohen-Macaulay module (Brennan-Herzog-Ulrich, 1987).

The name is motivated by the first appearance of these modules in a work by Ulrich (1984).

The existence of such modules over a given ring is in general an interesting problem. Indeed, Ulrich modules have useful applications in the study of associated graded rings, modules with linear resolutions, almost Gorenstein rings, and more.

One can specialize this definition to ideals, obtaining: an ideal I of R is an **Ulrich ideal** if

- $I^2 = QI$ where Q is a minimal reduction of I (stability condition),
- I/I^2 is free as R/I -module,

(Goto-Ozeki-Takahashi-Yoshida-Watanabe, 2014).

A typical example of Ulrich ideal is the maximal ideal of a local ring with minimal multiplicity.

It turns out that higher syzygy modules of Ulrich ideals are Ulrich modules.

Ulrich ideals have been studied in several classes of rings in papers such as: (Goto-Ozeki-Takahashi-Yoshida-Watanabe, 2016), (Goto-Takahashi-Taniguchi, 2018), (Goto-Isobe-Kumashiro, 2019), (Isobe, 2021).

2. Ulrich ideals in numerical semigroups

A submonoid $S \subseteq (\mathbb{N}, +)$ is a numerical semigroup if

$$|\mathbb{N} \setminus S| < \infty.$$

The Frobenius number of S is $f(S) = \max(\mathbb{N} \setminus S)$. The conductor of S is $c = f(S) + 1$.

An ideal of S is a subset E such that $E + S \subseteq E$. Examples of ideals are the maximal ideal $M = S \setminus \{0\}$ and the conductor $C = \{c, c + 1, \dots\}$.

Let E be an ideal of a numerical semigroup S . Set $a = \min E$.

We say that E is an **Ulrich ideal** if:

- $2E = a + E$ (stability condition),
- $E \setminus (a + E) = \bigsqcup_{x \in E \setminus (E + M)} (S \setminus E) + x$.

Examples:

- Principal ideals ($E = a + S$) are Ulrich.
- M is Ulrich if and only if S is stable (S has maximal embedding dimension).
- C is Ulrich if and only if $S = \{0, e, 2e, \dots, ne, \rightarrow\}$.

A non-principal Ulrich ideal satisfies the condition: $E = (a + S) - E$, which is equivalent to E being a stable trace ideal ($E = E + (S - E)$).

We have $C = S - \mathbb{N} \subseteq S - (E - E) = S - (E - a) = (a + S) - E = E$. Hence, a non-principal Ulrich ideal must contain C , and there are only finitely many non-principal Ulrich ideals.

Numerical criterion: An ideal E is Ulrich if and only if it is stable and

$$a = |S \setminus E| \cdot |E \setminus (E + M)|.$$

Example: Let $S = \langle 3, 10, 11 \rangle = \{0, 3, 6, 9, 10, 11, \rightarrow\}$. The non-principal ideals of S containing C are $M, C, E = \{6, 9, \rightarrow\}$. All of them are Ulrich. To check it for E , observe that

$$6 + E = \{12, 15, \rightarrow\} = 2E, \quad 6 = |S \setminus E| \cdot |E \setminus (E + M)| = 2 \cdot 3.$$

Known results from previous works:

- For a symmetric numerical semigroup S , there is a 1-1 correspondence between non-principal Ulrich ideals and symmetric oversemigroups generated over S by exactly one element (we will explore this correspondence more in general for good semigroups).
- For an almost symmetric non-symmetric numerical semigroup, the only possible non-principal Ulrich ideal is M (we gave an alternative proof not based on homological methods).

3. Value semigroups and good semigroups

Algebroid branch: one-dimensional domain of the form

$$R = \frac{k[[x_1, \dots, x_n]]}{P} \quad (k \text{ algebraically closed}).$$

Monomial curve: $k[[S]] := k[[t^{g_1}, \dots, t^{g_\nu}]]$

it is a particular algebroid branch.

Algebroid curve: one-dimensional, reduced ring of the form

$$R = \frac{k[[x_1, \dots, x_n]]}{P_1 \cap \dots \cap P_d} \quad (P_i \text{ are primes of height } n - 1).$$

$$R_i = \frac{k[[x_1, \dots, x_n]]}{P_i} \text{ is the } i\text{-th algebroid branch of } R.$$

$$Q(R) \cong k((t_1)) \times \dots \times k((t_d)) \text{ and } \overline{R} \cong k[[t_1]] \times \dots \times k[[t_d]].$$

Set $v(r) = (v_1(r_1), \dots, v_h(r_d))$ where v_i is the usual valuation on $k((t_i))$ corresponding to the ring $k[[t_i]]$.

The **value semigroup**:

$$S = v(R) := \{v(r) : r \in R, r \text{ non-zero divisor}\} \subset \mathbb{N}^d.$$

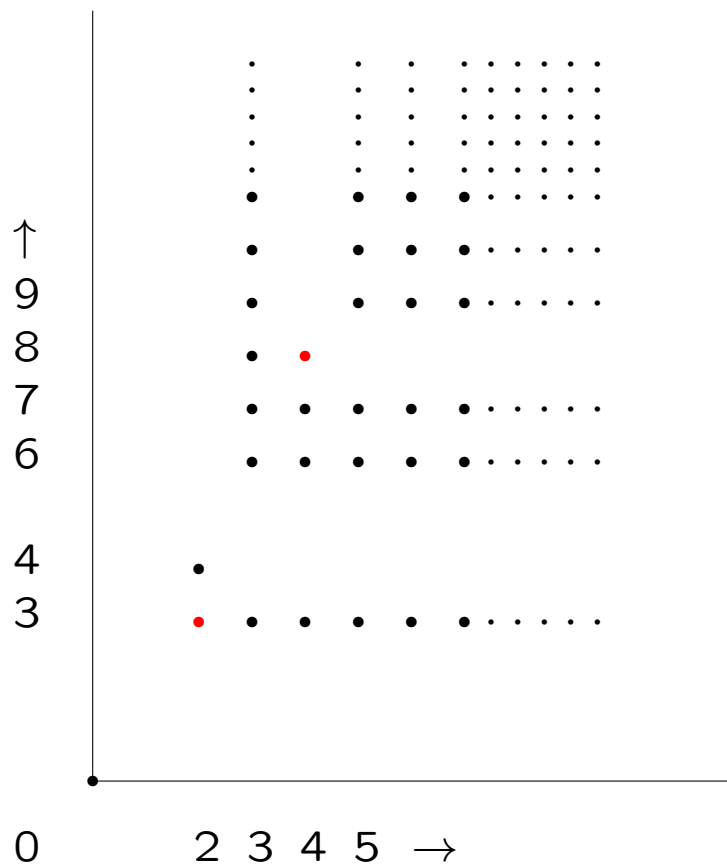
The value semigroup of an algebroid curve is a monoid that satisfies extra properties connected to valuations.

Given two elements $\alpha, \beta \in \mathbb{N}^d$ denote their infimum by

$$\alpha \wedge \beta := (\min(\alpha_1, \beta_1), \dots, \min(\alpha_d, \beta_d)).$$

The value semigroup of an algebroid curve satisfies the following properties:

- (G1) For every $\alpha, \beta \in S$, $\alpha \wedge \beta \in S$;
- (G2) Given two distinct elements $\alpha, \beta \in S$ such that $\alpha_i = \beta_i$, there exists $\epsilon \in S$ such that $\epsilon_i > \alpha_i = \beta_i$ and $\epsilon_j \geq \min\{\alpha_j, \beta_j\}$ for each $j \neq i$ (and if $\alpha_j \neq \beta_j$ the equality holds).
- (G3) There exists an element $c \in S$ such that $c + \mathbb{N}^d \subseteq S$. The minimal c satisfying this condition is called the **conductor** of S .
- (G4) $\alpha_i = 0$ for some i if and only if $\alpha = 0$.



$$R = \frac{k[[x,y,z]]}{(x^3-z^2) \cap (x^3-y^4)}$$

$$x \mapsto (t^2, u^4)$$

$$y \mapsto (0, u^3)$$

$$z \mapsto (t^3, 0)$$

$$v(x+y) = (2, 3)$$

$$e(R) = 5 = 2 + 3$$

$$\gamma = (4, 8), c = \gamma + (1, 1).$$

Picture 1. $S = v(R)$

A submonoid S of $\mathbb{N}^d$ satisfying properties (G1), (G2), (G3) is called a **good semigroup**. If moreover (G4) holds, S is **local**.

A semigroup ideal E satisfying (G1), (G2), (G3) is called a **good ideal**.

Non-local good semigroups arise naturally as value semigroups of certain blow-ups of algebroid curves.

Good semigroups are less well-behaved than numerical semigroups, indeed:

- good semigroups are not finitely generated as semigroups,
- operations on good ideals do not produce always good ideals,
- not all good semigroups arise as value semigroups of algebroid curves (Barucci-D'Anna-Fröberg, 2000).

4. Ulrich ideals of good semigroups

Let E be a good ideal of a good semigroup, and let a be the minimal element of E (which is well-defined by (G1)).

As in the numerical case, we say that E is an **Ulrich ideal** if:

- $2E = a + E$ (stability condition),
- $E \setminus (a + E) = \bigsqcup_{x \in E \setminus (E+M)} (S \setminus E) + x$.

Again we have that:

- principal ideals are Ulrich,
- non-principal Ulrich ideals must contain the conductor ideal $C = c + \mathbb{N}^d$.
- The maximal ideal $M = S \setminus \{0\}$ is Ulrich if and only if it is stable.

The main problem now is that the sets $S \setminus E$, $E \setminus (E + M)$ may be infinite.

For $\alpha, \beta \in S$, we say that $\alpha \ll \beta$ if $\alpha_i < \beta_i$ for every i .

Proposition: Let E be an Ulrich ideal of a good semigroup. Then:

- $E + M$ is a good ideal.
- Exactly one of the set $S \setminus E$, $E \setminus (E + M)$ is finite and the other one is infinite.
- If $S \setminus E$ is finite, then its elements are comparable with respect to the order $\ll$ to any other element of S .
- If $E \setminus (E + M)$ is finite, then its elements are comparable with respect to the order $\ll$ to any other element of E .

The main question to consider now is how to generalize the formula $a = |S \setminus E| \cdot |E \setminus (E + M)|$ holding in the numerical case.

We shall use the notion of distance. For a good ideal E , denote its minimal element by $\mathbf{a}_E$ and its conductor element by $\mathbf{c}_E$.

Given $\alpha, \beta \in E$ with $\alpha < \beta$, a saturated chain in E between α and β is a sequence

$$\alpha = \alpha^{(0)} < \alpha^{(1)} < \alpha^{(2)} < \dots < \alpha^{(p)} = \beta,$$

such that, for every i , $\alpha^{(i)}, \alpha^{(i+1)}$ are consecutive elements of E with respect to the standard partial order $\leq$.

All the saturated chains between two fixed elements of a good ideal have the same length (D'Anna, 1997). This length is the distance between α and β and it is denoted by $d_E(\alpha, \beta)$.

Given two good ideals $E \supseteq F$, define the distance

$$d(E \setminus F) = d_E(\mathbf{a}_E, \mathbf{c}_F) - d_F(\mathbf{a}_F, \mathbf{c}_F).$$

We have:

- $d(E \setminus F) = d(S \setminus F) - d(S \setminus E),$
- If $\mathbf{a} = (a_1, \dots, a_d)$, then $d(E \setminus (E + \mathbf{a})) = a_1 + \dots + a_d.$

Suppose now E to be Ulrich with minimal element $\mathbf{a}$. Then:

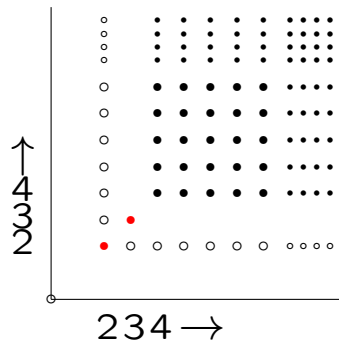
- If $S \setminus E$ is finite, then $d(S \setminus E) = |S \setminus E|$.
- Analogously, since $E + M$ is good, if $E \setminus (E + M)$ is finite, then $d(E \setminus (E + M)) = |E \setminus (E + M)|$.

Question: is E an Ulrich ideal if and only if it is stable and $\sum_{i=1}^d a_i = d(S \setminus E) \cdot d(E \setminus (E + M))$?

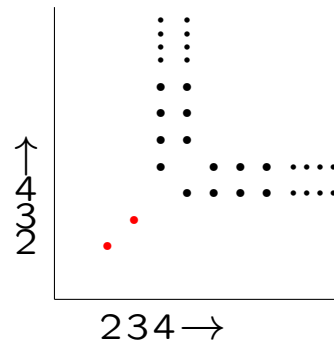
Theorem: Let $S \subseteq \mathbb{N}^2$ be a good semigroup and E an Ulrich ideal of S with minimal element $\mathbf{a} = (a_1, a_2)$. Then,

$$a_1 + a_2 \leq d(S \setminus E) \cdot d(E \setminus (E + M)).$$

Moreover, if the numerical projections of E are also Ulrich ideals in the respective projections of S , then equality holds in the above formula.



Picture 3.



Picture 4.

In Picture 3, we denote with circles the elements of $S \setminus E$ and with black and red dots the elements of E . The red dots are the elements of $E \setminus (E + M)$.

In Picture 4, we depict only the elements of $E \setminus 2E = E \setminus ((2, 2) + E)$. The red dots are the elements of $E \setminus (E + M)$.

5. Arf good semigroups

A good ideal of a good semigroup S is **integrally closed** if it contains all the elements of S larger than its minimal element.

A good semigroup S is **Arf** if for every $\alpha, \beta, \delta \in S$ with $\delta < \alpha, \beta$, we have $\alpha + \beta - \delta \in S$.

Theorem: Let $S \subseteq \mathbb{N}^d$ be an Arf good semigroup and let E be a non-principal ideal containing C . Then:

- If $d = 1$ and $S = \{0, s_1 = e, \dots, s_{n-1}, s_n, \rightarrow\}$, E is Ulrich if and only if it is integrally closed and its minimum element is $a = s_i = i \cdot e$.
- If $d > 1$ and $a = (a_1, \dots, a_d) = \min E$, E is Ulrich if and only if it is stable, $|S \setminus E| = n < \infty$, and $a = ne$. In the case E is Ulrich, $S \setminus E = \{0, e, \dots, (n-1)e\}$.
- E is Ulrich if and only if it is stable and $\sum_{i=1}^d a_i = d(S \setminus E) \cdot d(E \setminus (E + M))$.

6. Symmetric good semigroups

A numerical semigroup S is **symmetric** if, for every $s \in \mathbb{Z}$, $s \in S$ if and only if $f(S) - s \notin S$.

For an element $\alpha \in \mathbb{Z}^d$, set $\Delta(\alpha) = \bigcup_{i=1}^d \{\beta \in \mathbb{Z}^d : \beta_i = \alpha_i, \beta_j > \alpha_j \text{ for } j \neq i\}$.

Set $\gamma = c - (1, \dots, 1)$. A good semigroup $S \subseteq \mathbb{N}^d$ is **symmetric** if, for every $\alpha \in \mathbb{Z}^d$, $\alpha \in S$ if and only if $\Delta(\gamma - \alpha) \cap S = \emptyset$.

Theorem: Let $S \subseteq \mathbb{N}^d$ be a symmetric good semigroup and let E be a non-principal Ulrich ideal with minimal element a . Then:

- Either E is generated by two elements a, b or is generated by a and $\Delta(b)$ for some $b \in a + S$.
- $\sum_{i=1}^d a_i = d(S \setminus E) \cdot d(E \setminus (E + M)) = 2d(S \setminus E)$.

A **relative ideal** of a good semigroups S is a subset E of $\mathbb{N}^d$ such that, for some $\alpha \in S$, $\alpha + E$ is an ideal of S .

Theorem: Let $S \subseteq \mathbb{N}^d$ be a symmetric good semigroup. There is a 1-1 correspondence between:

- the Ulrich ideals of S generated by two elements.
- the symmetric semigroups T such that $S \subseteq T \subseteq \mathbb{N}^d$ and T is generated by two elements $0, x$ as a relative ideal of S .

This correspondence is defined explicitly by the mapping

$$E \rightarrow E - E, \quad T \rightarrow S - T.$$

7. One-dimensional Cohen-Macaulay rings

Let R be a one-dimensional local analytically unramified Cohen-Macaulay ring, with infinite residue field.

Let S be the value semigroup of R . S is a good semigroup contained in $\mathbb{N}^d$, where d is the number of minimal primes of R .

Theorem: Let I be a stable ideal of R ($I^2 = xI$ for some regular $x \in I$). Set $E = v(I)$ and let $\mathbf{a} = (a_1, \dots, a_d) = \min(E)$.

- If $\sum_{i=1}^d a_i = d(S \setminus E) \cdot d(E \setminus (E + M))$, then I is an Ulrich ideal.
- If I is Ulrich and $\nu(I) = d(E \setminus E + M)$, then $\sum_{i=1}^d a_i = d(S \setminus E) \cdot d(E \setminus (E + M))$.
- If R is Gorenstein and E is Ulrich, then I is Ulrich.
- If R is an Arf ring, then I is Ulrich if and only if E is an Ulrich ideal.

Let S be a numerical semigroup.

Recall that the type $t(S)$ of S is the cardinality $|(S - M) \setminus S|$. We have:

Proposition: Let E be an Ulrich ideal of a numerical semigroup S . Then, its number of generators is

$$\nu(E) = \frac{|(E - M) \setminus E|}{|(E -_S M) \setminus E|}.$$

Moreover, $t(S) \geq (\nu(E) - 1)|(E -_S M) \setminus E|$.

Let I be a regular ideal of a one-dimensional local Cohen-Macaulay ring R , and let $x \in I$ be a regular element.

The type $t(I)$ is the dimension as an $R/\mathfrak{m}$ -vector space of the module $(0) :_{I/xI} \mathfrak{m} \cong (xI :_I \mathfrak{m})/xI \cong (I :_{Q(R)} \mathfrak{m})/I$.

Proposition: Let I be an Ulrich ideal of a one-dimensional local Cohen-Macaulay ring R . Then $\nu(I)t(R/I) = t(I)$.

THANKS FOR YOUR ATTENTION!